

FSISIM Acronym & Term Glossary

Aviation Acronyms (APU, FMC, FMS, IRS, ILS, RA, EICAS, ECAM, VNAV, LNAV)

This glossary section defines critical aviation acronyms commonly encountered during simulator operations and flight deck procedures. Familiarity with these terms is essential for effective crew resource management and proper system operation.

APU (Auxiliary Power Unit) - A self-contained gas turbine engine typically located in the tail cone that provides electrical power and pneumatic air for air conditioning and engine starting when main engines are not operating. APU bleed air can also pressurize hydraulic reservoirs on some aircraft types.

FMC (Flight Management Computer) - The computational core of the flight management system that performs navigation calculations, performance optimization, and flight plan management. The FMC interfaces with the CDU (Control Display Unit) for crew input and displays.

FMS (Flight Management System) - An integrated avionics suite combining navigation, flight planning, and aircraft performance management. The FMS typically includes the FMC, CDU, navigation database, and associated sensors (IRS, GPS, radio navigation aids).

IRS (Inertial Reference System) - A self-contained navigation system using laser gyroscopes or ring laser gyros to determine aircraft position, velocity, attitude, and heading through dead reckoning. Modern aircraft typically employ three independent IRS units for redundancy.

ILS (Instrument Landing System) - A ground-based precision approach aid transmitting localizer (lateral guidance) and glideslope (vertical guidance) signals. Standard ILS categories range from CAT I (200-foot DH) through CAT IIIc (zero visibility operations).

RA (Radio Altimeter) - A radar-based system measuring absolute height above terrain by transmitting radio frequency signals and measuring return time. RA provides critical input for GPWS/EGPWS alerting and autoland operations below 2,500 feet AGL.

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EICAS (Engine Indication and Crew Alerting System) - Boeing's integrated display system presenting engine parameters, crew alerts, and system status messages on primary and secondary display screens. Alert hierarchy follows a color-coded priority system:

1. **Red (Warning)** - Immediate crew action required
2. **Amber (Caution)** - Timely crew awareness/action needed
3. **Blue (Advisory)** - Crew awareness of system status
4. **White (Memo)** - Operational information

ECAM (Electronic Centralized Aircraft Monitor) - Airbus's equivalent to EICAS, displaying system synoptics, procedures, and alerts on upper and lower display units.

VNAV (Vertical Navigation) - FMS-managed vertical flight path control maintaining programmed altitude constraints and optimized climb/descent profiles.

LNAV (Lateral Navigation) - FMS-managed lateral flight path control following programmed route waypoints and procedures.

Simulator Hardware Terms (IOS, motion platform, control loading, visual system)

Simulator Hardware Terms

The FSISIM platform integrates multiple hardware subsystems to provide high-fidelity flight training. Understanding these components is essential for effective operation and troubleshooting.

Instructor Operating Station (IOS) - The primary interface for simulator control and scenario management. The IOS workstation provides real-time aircraft state monitoring, environmental control, malfunction insertion, and session recording capabilities. Modern IOS configurations typically feature dual or triple monitor arrays with touchscreen capability for rapid access to training functions.

Motion Platform - A six degree-of-freedom (6-DOF) electric or hydraulic actuator system that replicates aircraft movement. The motion cueing algorithm translates flight dynamics into platform motion within the physical constraints of actuator stroke limits (± 24 inches typical). Motion washout filters prevent platform saturation during sustained accelerations while maintaining pilot perception fidelity. Platform BIT sequences verify actuator response, load cell calibration, and emergency stop functionality.

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Control Loading System - Force feedback mechanisms applied to flight controls to simulate aerodynamic forces. The system employs servo motors or hydraulic actuators with real-time force computation based on airspeed, control surface deflection, and configuration. Typical breakout forces range from 2-15 pounds for yoke/sidestick and 50-150 pounds for rudder pedals. Control loading profiles are validated against flight test data and updated via FMS database revisions.

Visual System - High-resolution projection or LED display array providing outside world imagery. Current generation systems deliver 200° horizontal by 40° vertical field-of-view minimum, with pixel density exceeding 30 arc-minutes per pixel. The image generator (IG) renders terrain databases at 60Hz minimum refresh rate with real-time lighting, weather effects, and airport detail. Visual system alignment procedures include:

1. Execute visual alignment BIT from IOS maintenance menu
2. Verify projector lamp hours and color balance
3. Confirm edge-blending calibration across display segments
4. Validate geometric correction tables for zero distortion
5. Test night lighting ratios and contrast settings

Database correlation between visual, navigation, and terrain awareness systems ensures consistent environmental representation across all simulator subsystems.

Maintenance Terms (BIT, fault code, MMI, lost time categories)

Built-In Test (BIT) systems provide automated diagnostic capabilities for simulator hardware and software components. The FSISIM platform implements continuous BIT (CBIT) monitoring during operation and initiated BIT (IBIT) sequences during power-up or maintenance procedures. BIT results are logged in the Maintenance Management Interface (MMI) with timestamp, fault code, and affected Line Replaceable Unit (LRU) identification. Successful BIT completion generates a code 0000, while any detected anomaly produces a specific four-digit hexadecimal fault code for troubleshooting reference.

Fault codes in the FSISIM architecture follow ARINC 624 standards with manufacturer-specific extensions. Codes are categorized by severity level: Level 1 (advisory), Level 2 (caution requiring attention before next session), and Level 3 (warning requiring immediate corrective action). The MMI displays fault codes with associated Fault Isolation Manual (FIM) references and recommended corrective actions. Intermittent faults are tracked through occurrence counters, with automatic escalation after three instances within a 72-hour period.

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Maintenance Management Interface (MMI) serves as the primary diagnostic and configuration tool for simulator technicians. The MMI provides real-time system status monitoring, fault history retrieval, software version control, and configuration database management. Access requires technician-level authentication credentials. Key MMI functions include:

- System health monitoring with graphical status displays
- Fault code retrieval and historical trending analysis
- LRU configuration and calibration procedures
- Software load verification and update management
- Preventive maintenance scheduling and tracking
- Session replay for fault reproduction

Lost time categories classify simulator unavailability for training operations. Category A represents scheduled maintenance with advance notice (no impact on training availability metrics). Category B indicates unscheduled maintenance completed within the same training day. Category C designates extended downtime exceeding one training day, requiring regulatory reporting per applicable qualification standards. Mean Time Between Failures (MTBF) and Mean Time To Repair (MTTR) calculations exclude Category A events but incorporate Categories B and C for availability performance tracking.

Note Type Codes (INI, UPD, SOL, RES)

Flight simulator technical documentation employs standardized note type codes to categorize maintenance entries, system annotations, and operational advisories. These four-character codes appear in system logs, maintenance action forms (MAFs), and configuration management databases to ensure consistent tracking and resolution of simulator discrepancies.

INI (Initial) designates the first documentation of a newly discovered discrepancy, malfunction, or configuration variance. When a technician or instructor identifies an issue—such as an FMC database mismatch, IRS alignment drift beyond tolerance, or EICAS display anomaly—the initial entry receives an INI code. This triggers the fault tracking workflow and assigns a unique reference number for subsequent actions. INI entries must include fault code identification, affected LRU (Line Replaceable Unit), and initial troubleshooting results from BIT (Built-In Test) sequences.

UPD (Update) codes document ongoing investigation, interim corrective actions, or status changes for open discrepancies. Examples include partial repairs requiring additional parts,

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software patches awaiting validation, or deferred maintenance items. An UPD entry might note: "APU start sequence fault—replaced starter relay per AMM 49-21-00, monitoring for recurrence over next 10 flight cycles."

SOL (Solution) indicates successful resolution of the documented issue. The entry must reference the original INI code, describe corrective actions taken, and confirm system functionality through operational checks. For instance: "ILS localizer deviation corrected—recalibrated antenna alignment per station survey data, verified approach coupling within ± 0.5 dot deviation."

RES (Reset) applies when discrepancies resolve without physical intervention, typically following software resets, database updates, or environmental condition changes. Common RES scenarios include:

- Transient BITE faults clearing after power cycle
- VNAV path computation errors resolved via FMS database reload
- ECAM advisory messages extinguishing after system warm-up

All note type codes require technician signature, timestamp, and cross-reference to applicable maintenance manuals or service bulletins for audit trail integrity.